

Digital Logic Assignment Solutions

Assignment 1

October 16, 2025

1 Number Conversions

1.1 Convert 123.4 to base 7, base 12, and base 16

a) $(123.4)_{10}$ in base 7

Solution:

Take integer and repeat the division.

Iterations	Quotient	Remainder	Coefficient
123/7	17	4	$a_0 = 4$
17/7	2	3	$a_1 = 3$
2/7	0	2	$a_2 = 2$

Thus the integer part is $(234)_7$

Take fraction part and repeat the multiplication.

Iterations	Integer	Fraction	Coefficient
$0.4*7$	2	.8	$a_{-1} = 2$
$0.8*7$	5	.6	$a_{-2} = 5$

Thus the fraction part is $(.25)_7$

So the final answer should be $(234.25)_7$.

b) $(123.4)_{10}$ in base 12

Solution:

Take integer and repeat the division.

Iterations	Quotient	Remainder	Coefficient
123/12	10	3	$a_0 = 3$
10/12	0	10 (A)	$a_1 = A$

Thus the integer part is $(A3)_{12}$

Take fraction part and repeat the multiplication.

Iterations	Integer	Fraction	Coefficient
$0.4*12$	4	.8	$a_{-1} = 4$
$0.8*12$	9	.6	$a_{-2} = 9$

Thus the fraction part is $(.49)_{12}$

So the final answer should be $(A3.49)_{12}$.

c) $(123.4)_{10}$ in base 16

Solution:

Take integer and repeat the division.

Iterations	Quotient	Remainder	Coefficient
123/16	7	11 (B)	$a_0 = B$
7/16	0	7	$a_1 = 7$

Thus the integer part is $(7B)_{16}$

Take fraction part and repeat the multiplication.

Iterations	Integer	Fraction	Coefficient
0.4×16	6	.4	$a_{-1} = 6$
0.4×16	6	.4	$a_{-2} = 6$

Thus the fraction part is $(.66)_{16}$

So the final answer should be $(7B.66)_{16}$.

2 Subtraction using the 10's complement

2.1 (a) 4637 - 2579

2579 10's complement = 7421

$$\begin{array}{r}
 4637 \\
 + 7421 \quad (10\text{'s complement of } 2579) \\
 \hline
 12058 \\
 - 10000 \\
 \hline
 2058
 \end{array}$$

Result: 2058

2.2 (b) 125 - 1800

1800 10's complement = 8200

$$\begin{array}{r}
 125 \\
 + 8200 \quad (10\text{'s complement of } 1800) \\
 \hline
 8325
 \end{array}$$

8325 10's complement = 1675

Result: -1675

3 Boolean Expressions Simplification

3.1 $yz'(z + xz') + (x' + z')(x'y + x'z)$

$$\begin{aligned}
 &= xyz' + x'y + x'z + x'yz' \\
 &= yz' + x'y + x'z \\
 &= yz' + x'yz' + x'yz + x'z \\
 &= (1 + x')yz' + (y + 1)x'z \\
 &= yz' + x'z
 \end{aligned}$$

3.2 $A(B + C) + BD'(A' + C)$

$$\begin{aligned}
 &= AB + AC + A'BD' + BCD' \\
 &= B(A + A'D') + AC + BCD' \\
 &= B(A + D') + AC + BCD' \\
 &= AB + AC + BD'(1 + C) \\
 &= A(B + C) + BD'
 \end{aligned}$$

4 Express the Boolean expression in sum of minterms form

4.1 $bd' + acd' + ab'c + a'c'$

$$\begin{aligned}
 &= bd'(c + c')(a + a') + a(b + b')cd' + ab'c(d + d') + a'(b + b')c'(d + d') \\
 &= a'b'c'd' + a'b'c'd + a'bc'd' + a'bc'd + a'bcd' + ab'cd' + ab'cd + abc'd' + abcd' \\
 &= \Sigma(0, 1, 4, 5, 6, 10, 11, 12, 14)
 \end{aligned}$$

5 Express the Boolean expression in product of maxterms form

5.1 $a + b'c$

$$\begin{aligned}
 &= (a + b')(a + c) \\
 &= (a + b' + cc')(a + bb' + c) \\
 &= (a + b' + c)(a + b' + c')(a + b + c)(a + b' + c) \\
 &= \prod(0, 2, 3)
 \end{aligned}$$

6 Three-variable Boolean Functions Algebraically

6.1 $F1(A, B, C) = \Sigma(0, 1, 2, 3, 5)$ to sum of product form

From F1 get the truth table:

$A \setminus BC$	00	01	11	10
0	1	1	1	1
1	0	1	0	0

$$\begin{aligned}
 F1 &= \Sigma(0, 1, 2, 3, 5) \\
 &= A' + B'C
 \end{aligned}$$

6.2 $F2(B, C, D) = (cd + b'c + bd')(b + d)$ into product of sum form

The SOM form:

$$\begin{aligned}
 F2 &= BCD + CD + BD' + B'CD \\
 &= \Sigma(3, 4, 6, 7)
 \end{aligned}$$

Get the K-map with the SOM form:

$B \setminus CD$	00	01	11	10
0	0	0	1	0
1	1	0	1	1

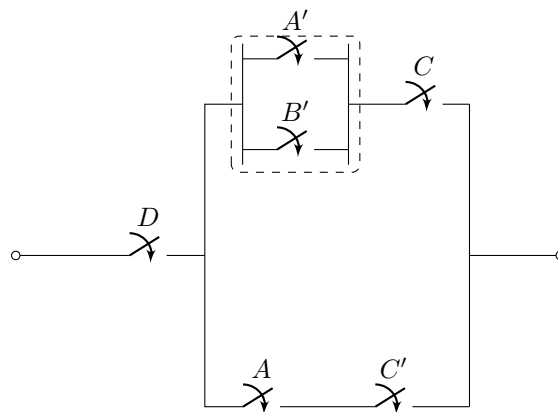
The SOP expression:

$$\begin{aligned} F2' &= \Sigma(0, 1, 2, 5) \\ &= B'D' + C'D \end{aligned}$$

Get the POS expression with using demorgan's law:

$$\begin{aligned} F2 &= (F2')' \\ &= (B'D' + C'D)' \\ &= (B + D)(C + D') \end{aligned}$$

7 Switch Circuit



7.1 Derive the switching algebra expression

Simplify the following expression:

$$F(A, B, C, D) = D((A' + B')C + AC')$$

7.2 Derive the truth table

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

7.3 Simplify the algebra expression

$AB \backslash CD$	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	0	1	0	0
10	0	1	1	0

From the truth table:

$$\begin{aligned}
 F &= A'CD + AC'D + AB'D \\
 &= D(A'C + AC' + AB') \\
 &= D(A'C + A(B' + C'))
 \end{aligned}$$

8 Draw the circuit

